

2024 AMC 10A Problem 23 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 23

Problem:

Integers a , b , and c satisfy $ab + c = 100$, $bc + a = 87$, and $ca + b = 60$. What is $ab + bc + ca$?

Answer Choices:

A. 212

B. 247

C. 258

D. 276

E. 284

Solution:

Notice that the difference between 100 and 87 is 13, a prime number. This fact will help to simplify the problem. Subtract the second equation from the first to get

$$\begin{aligned} 13 &= (ab + c) - (bc + a) \\ &= ab - bc - a + c \\ &= b(a - c) - (a - c) \\ &= (b - 1)(a - c) \end{aligned}$$

Thus $b - 1 = \pm 1$ or $b - 1 = \pm 13$.

- If $b - 1 = -1$, then $b = 0$, implying $c = 100$, $a = 87$, and $ca = 60$, which is impossible.
- If $b - 1 = 1$, then $b = 2$ and $a - c = 13$, implying $ca = 58 = 2 \cdot 29$, which cannot be true if $a - c = 13$.
- If $b - 1 = 13$, then $b = 14$ and $a - c = 1$, implying $ca = 46 = 2 \cdot 23$, which cannot be true if $a - c = 1$.
- If $b - 1 = -13$, then $b = -12$ and $a - c = -1$, implying $ca = 72$, which is satisfied when $a = -9$ and $c = -8$. In fact, $a = -9$, $b = -12$, and $c = -8$ satisfies all three equations.

The requested value is $ab + bc + ca = (-9)(-12) + (-12)(-8) + (-8)(-9) = 108 + 96 + 72 = (D) \boxed{276}$.

Note: There are also four noninteger solutions. When written in the form (a, b, c) , these solutions are approximately $(0.594, 0.869, 99.484)$, $(1.715, 57.455, 1.484)$, $(7.477, 12.525, 6.349)$, and $(86.214, 1.152, 0.683)$.

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